R18

Code No: 152AE

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year II Semester Examinations, July/August - 2021
APPLIED PHYSICS
(Common to EEE, CSE, IT, ITE)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a)	Explain a black body radiation with example.	
b)	Derive an expression for Particle in one dimensional box.	[7+8]
2.a)	Differentiate between Intrinsic and Extrinsic semiconductors.	
b)	Write a note on Carrier generation and recombination.	[7+8]
3.a)	Explain the formation of Extrinsic semiconductors with diagram.	
b)	Explain the formation Bipolar junction transistor with its characteristics.	[7+8]
4.a)	Describe the recombination mechanism in non-radiative semiconductors.	
b)	Explain LED and its uses.	[8+7]
5.a)	Describe the recombination mechanism in radiative semiconductors.	
b)	What is a "LASER"? What are the merits of a semiconductor laser?	[7+8]
6.a)	Explain the principle and working of He-Ne laser.	
b)	Mention the various types of Losses in optical fibres.	[7+8]
7.a)	Explain permittivity and Dielectric constant for a magnetic material.	
b)	Derive Clausius-Mossotti Equation.	[7+8]
8.a)	How many types of magnetic materials with examples- Classify?	
b)	Mention eight applications of magnetic materials.	[7+8]

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